Exploring supernova neutrino mass ordering at DUNE via quantum entanglement

Adikiran Johny

Department of Physics
Central University of Karnataka

November 21, 2025

Three-Flavor Neutrino Oscillations

- Neutrino flavor eigenstates ν_α ($\alpha = e, \mu, \tau$) are related to the mass eigenstates ν_i ($i = 1, 2, 3$) through the unitary PMNS matrix:

$$|\nu_\alpha\rangle = \sum_i U_{\alpha i} |\nu_i\rangle . \tag{1}$$

- A standard parameterization of U is

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{13}s_{23}e^{i\delta} & c_{12}c_{23} - s_{12}s_{13}s_{23}e^{i\delta} & c_{13}s_{23} \\ s_{12}s_{23} - c_{12}s_{13}c_{23}e^{i\delta} & -c_{12}s_{23} - s_{12}s_{13}c_{23}e^{i\delta} & c_{13}c_{23} \end{pmatrix} . \tag{2}$$

  where $c_{ij} = \cos\theta_{ij}$, $s_{ij} = \sin\theta_{ij}$ and δ_{CP} denotes the CP-violating phase.

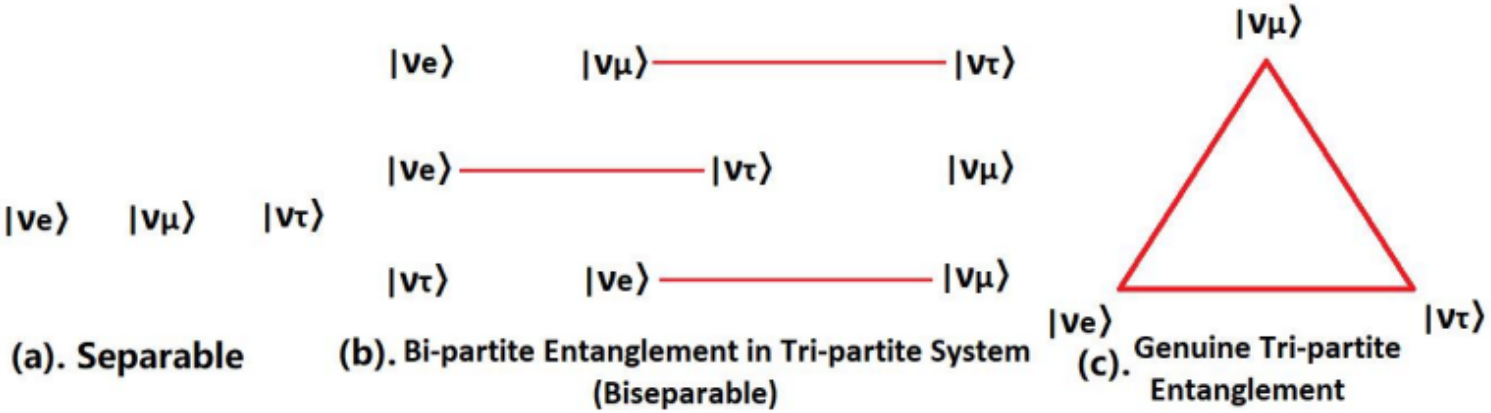
- Each mass eigenstate evolves as

$$|\nu_i(t)\rangle = e^{-iE_i t} |\nu_i\rangle , \tag{3}$$

- The time-dependent flavor state is

$$|\nu_\alpha(t)\rangle = \sum_i U_{\alpha i}\, e^{-iE_i t}\, |\nu_i\rangle . \tag{4}$$

- Three types of entanglement:



- For three flavors, the single-particle states can be represented as

$$
\begin{aligned}
|\nu_e\rangle &= |1\rangle_e \otimes |0\rangle_\mu \otimes |0\rangle_\tau \equiv |100\rangle_e \,, \\
|\nu_\mu\rangle &= |0\rangle_e \otimes |1\rangle_\mu \otimes |0\rangle_\tau \equiv |010\rangle_\mu \,, \\
|\nu_\tau\rangle &= |0\rangle_e \otimes |0\rangle_\mu \otimes |1\rangle_\tau \equiv |001\rangle_\tau \,.
\end{aligned}
$$

- The time-evolved flavor state reads

$$
|\nu_\alpha(t)\rangle = \bar{U}_{\alpha e}(t)\,|100\rangle_{e\mu\tau} + \bar{U}_{\alpha\mu}(t)\,|010\rangle_{e\mu\tau} + \bar{U}_{\alpha\tau}(t)\,|001\rangle_{e\mu\tau} \,, \qquad (5)
$$

with $\bar{U} = U e^{-i\mathcal{H}_m t} U^\dagger$ and $\mathcal{H}_m = \mathrm{diag}(E_1, E_2, E_3)$.

Density Matrix

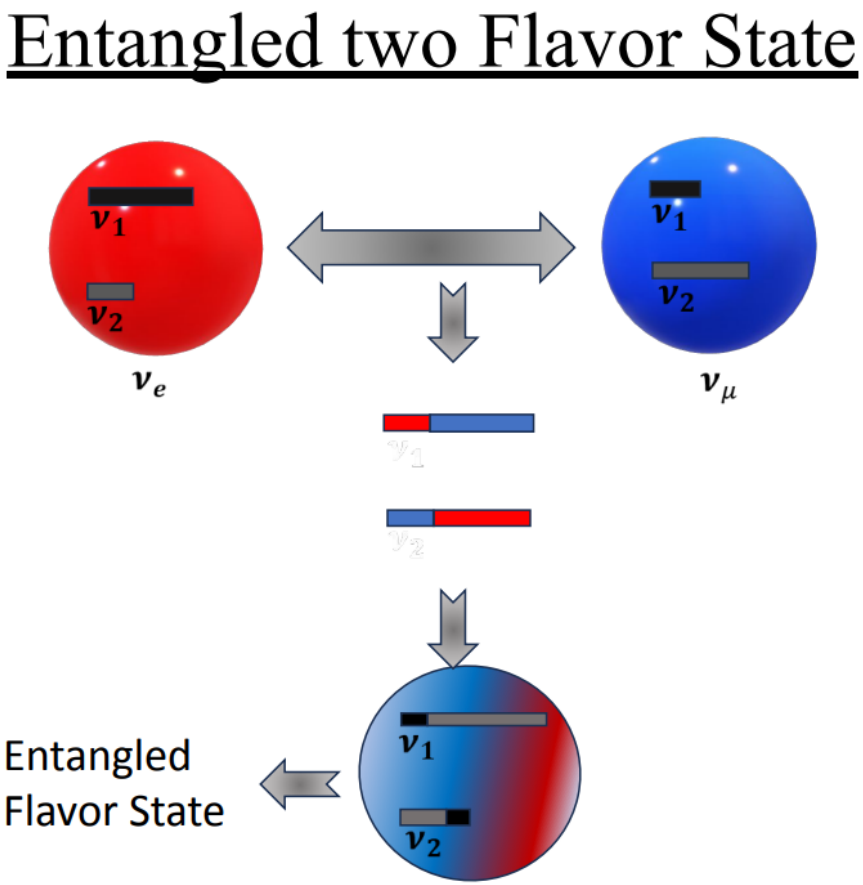
- The corresponding density matrix is

$$\rho^{\alpha}_{ABC}(t) = |\nu_\alpha(t)\rangle \langle \nu_\alpha(t)| = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \rho^{\alpha}_{22} & \rho^{\alpha}_{23} & 0 & \rho^{\alpha}_{25} & 0 & 0 & 0 \\ 0 & \rho^{\alpha}_{32} & \rho^{\alpha}_{33} & 0 & \rho^{\alpha}_{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \rho^{\alpha}_{52} & \rho^{\alpha}_{53} & 0 & \rho^{\alpha}_{55} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \tag{6}$$

with nonzero entries

$$\begin{aligned} \rho^{\alpha}_{22} &= |\bar{U}_{\alpha\tau}(t)|^2, \quad \rho^{\alpha}_{33} = |\bar{U}_{\alpha\mu}(t)|^2, \quad \rho^{\alpha}_{55} = |\bar{U}_{\alpha e}(t)|^2, \\ \rho^{\alpha}_{23} &= \bar{U}_{\alpha\tau}(t)\bar{U}^*_{\alpha\mu}(t), \quad \rho^{\alpha}_{25} = \bar{U}_{\alpha\tau}(t)\bar{U}^*_{\alpha e}(t), \\ \rho^{\alpha}_{35} &= \bar{U}_{\alpha\mu}(t)\bar{U}^*_{\alpha e}(t), \end{aligned} \tag{7}$$

Bi-partite entanglement in neutrino oscillations



1. Consider a pure state of electron(red) and muon(blue) neutrino with superimposed mass eigenstate as ν_1(black) and ν_2(grey).

$$|\nu_\beta\rangle = -\sin\theta\,|\nu_1\rangle + \cos\theta\,|\nu_2\rangle$$

$$|\nu_\alpha\rangle = \cos\theta\,|\nu_1\rangle + \sin\theta\,|\nu_2\rangle$$

2. While propagating the mass eigenstate acquires some flavor state by the rotation of basis.

$$|\nu_e\rangle = |0\rangle_e \otimes |1\rangle_\mu = |01\rangle_e$$
$$|\nu_\mu\rangle = |1\rangle_e \otimes |\nu\rangle_\mu = |10\rangle_\mu$$

3. Therefore, the superimposed mass eigenstate associated with each flavor basis now gained some flavor state with it, which implies the state is no longer pure.

$$|\nu_e(\theta,\phi)\rangle = (\cos^2\theta + \sin^2\theta e^{-i\phi})\,|\nu_e\rangle - \sin\theta\cos\theta(1 - e^{-i\phi})\,|\nu_\mu\rangle$$

[1]

[1] Credit: S. Patra, IIT Bhilai

Measurement of Entanglement

Neutrinos entanglement has been explored in various ways such as: [2]

- **Entanglement of Formation (EOF):** EOF of a bipartite state ρ is the minimum average number of Bell pairs (e-bits) needed to prepare using only local operations and classical communication.
- **Concurrence :** A quantitative measure of entanglement for a bipartite system.
- **Negativity :** A quantitative measure of quantum entanglement between two subsystems of a composite quantum state.

[2] L. Konwar1 and B. Yadav, arXiv:2402.09952[hep-ph].

Entanglement of Formation

- For a pure tripartite system $\rho_{ABC}(t)$, Entanglement of Formation can be defined as

$$EOF(\rho_{ABC}(t)) = \frac{1}{2}\left[S(\rho_A) + S(\rho_B) + S(\rho_C)\right], \tag{8}$$

where,

$$S(\rho) = -\mathrm{Tr}(\rho \log_2 \rho)$$

denotes the von Neumann entropy with

$$\rho_A = \mathrm{Tr}_{BC}(\rho_{ABC}(t)), \rho_B = \mathrm{Tr}_{AC}(\rho_{ABC}(t)), \rho_C = \mathrm{Tr}_{AB}(\rho_{ABC}(t)),$$

are the corresponding reduced density matrices.

- EOF in terms of survival and transition probabilities in vacuum :

$$\begin{aligned} EOF^{\alpha} = -\frac{1}{2}\Big[\quad & P_{\alpha e}\log_2 P_{\alpha e} + P_{\alpha\mu}\log_2 P_{\alpha\mu} + P_{\alpha\tau}\log_2 P_{\alpha\tau} \\ & +(P_{\alpha\mu} + P_{\alpha\tau})\log_2(P_{\alpha\mu} + P_{\alpha\tau}) \\ & +(P_{\alpha e} + P_{\alpha\tau})\log_2(P_{\alpha e} + P_{\alpha\tau}) \\ & +(P_{\alpha\mu} + P_{\alpha e})\log_2(P_{\alpha\mu} + P_{\alpha e})\Big], \end{aligned} \tag{9}$$

where $\alpha = e, \mu, \tau$ denotes the initial neutrino flavor state.

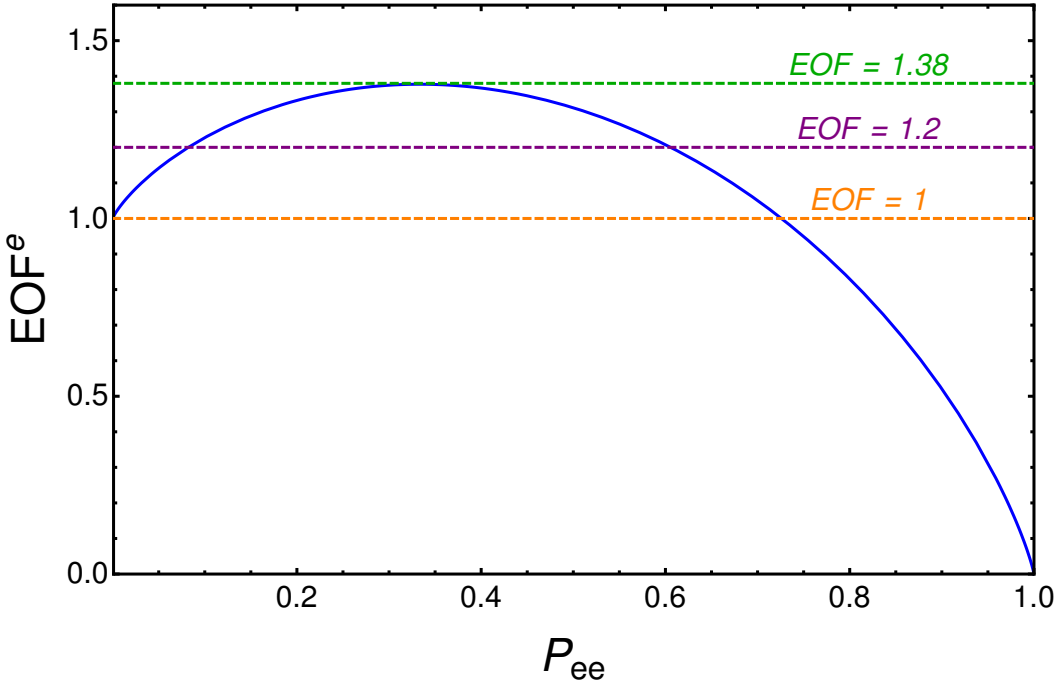


Figure: EOF with survival probability of electron neutrino.

Cases	P0	P1	P2	P3	P4
EOF	0.000	0.726	0.606	0.333	0.082

Table: Benchmark values of P_{ee} for EOF

Concurrence

- For a pure tripartite system,the generalized concurrence can be written as

$$C(\rho_{ABC}) = \left[3 - \mathrm{Tr}(\rho_A^2) - \mathrm{Tr}(\rho_B^2) - \mathrm{Tr}(\rho_C^2)\right]^{\frac{1}{2}}, \tag{10}$$

where

$$\rho_A = \mathrm{Tr}_{BC}(\rho_{ABC}(t)), \rho_B = \mathrm{Tr}_{AC}(\rho_{ABC}(t)), \text{ and } \rho_C = \mathrm{Tr}_{AB}(\rho_{ABC}(t))$$

- Concurrence in terms of survival and transition probabilities can be expressed as:

$$C^{\alpha} = \sqrt{3 - 3P_S - 2P_{\alpha\mu}P_{\alpha\tau} - 2P_{\alpha e}(P_{\alpha\mu} + P_{\alpha\tau})}, \tag{11}$$

where $P_S = P_{\alpha e}^2 + P_{\alpha\mu}^2 + P_{\alpha\tau}^2$ denotes the sum of squared probabilities.

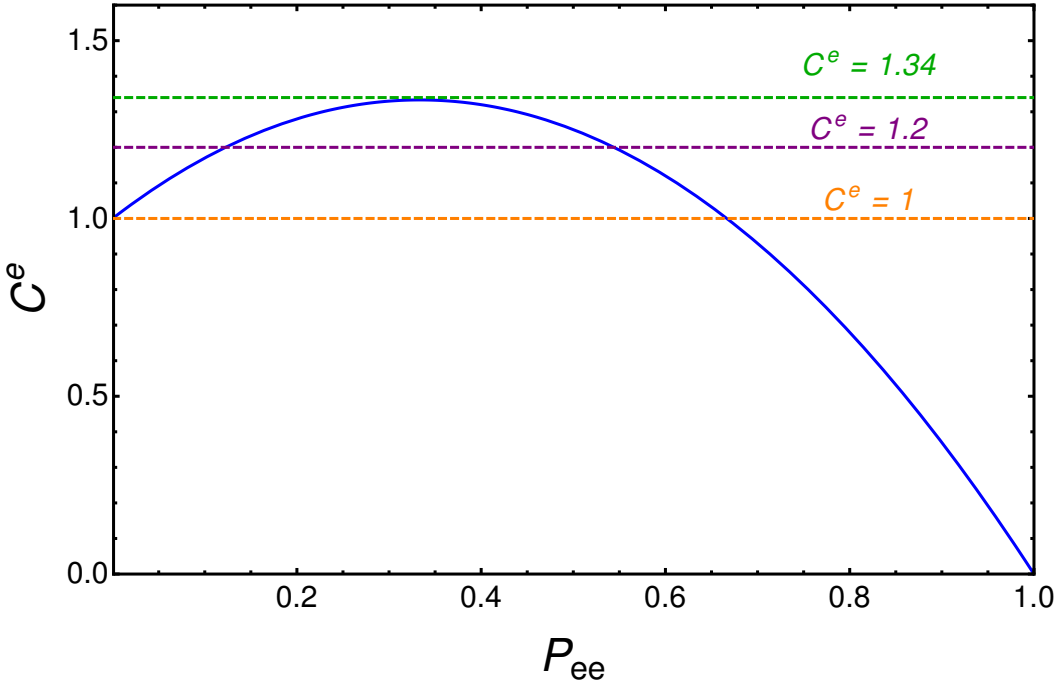


Figure: Concurrence with survival probability of electron neutrino.

Cases	P0	P1	P2	P3	P4
Concurrence	0.000	0.667	0.544	0.333	0.123

Table: Benchmark values of P_{ee} for Concurrence

Negativity

- For a three-qubit state, the negativity is defined as

$$N = (N_{A-BC} N_{B-CA} N_{C-AB})^{\frac{1}{3}}, \tag{12}$$

where,

$$N_{A-BC} = -\sum_i \lambda_i^A, N_{B-CA} = -\sum_j \lambda_j^B, \text{ and } N_{C-AB} = -\sum_k \lambda_k^C ,$$

with the quantities λ_i^A, λ_j^B, and λ_k^C represent the negative eigenvalues of the partially transposed density matrices $\rho_{ABC}^{T_\alpha}(t)$ with respect to each subsystem.

- Negativity in terms of survival and transition probabilities in vacuum

$$N^\alpha = \left[\sqrt{P_{\alpha e}}\sqrt{P_{\alpha\mu} + P_{\alpha\tau}}\sqrt{P_{\alpha e}}\sqrt{P_{\alpha\mu}}\sqrt{P_{\alpha e} + P_{\alpha\mu}}\sqrt{P_{\alpha\tau}}\right]^{\frac{1}{3}} . \tag{13}$$

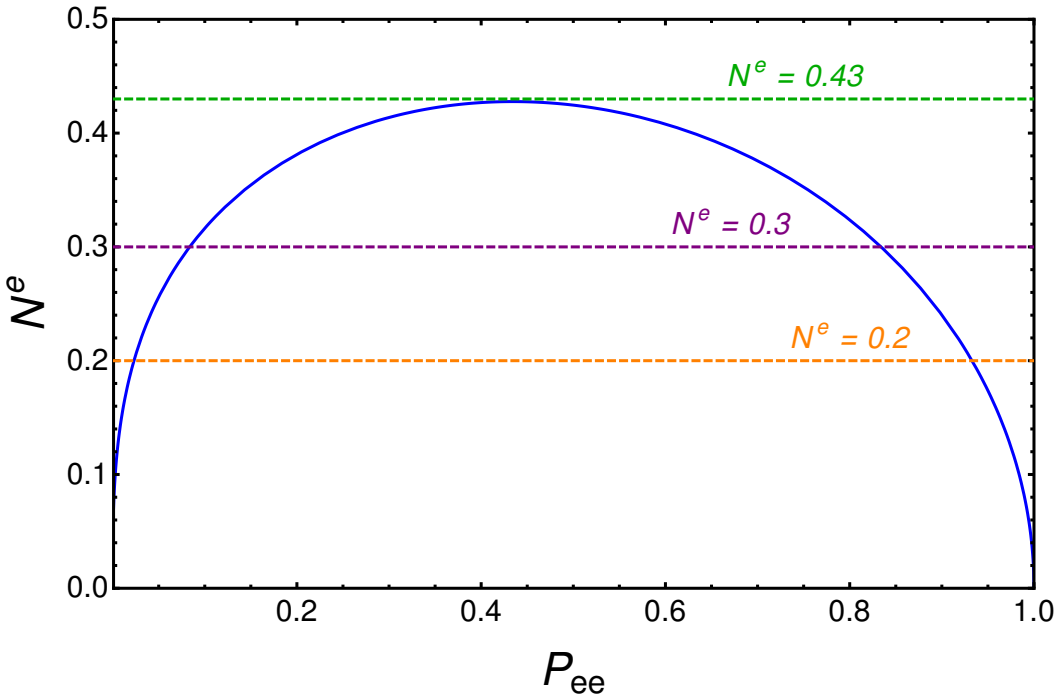


Figure: Negativity with survival probability of electron neutrino.

Cases	P0	P1	P2	P3	P4
Negativity	0.000	0.698	0.434	0.083	0.023

Table: Benchmark values of P_{ee} for Negativity

Experimental Set up and Simulation Details

- To simulate the detection of supernova neutrinos at DUNE, we employ the **SNOwGLoBES**[3] framework, which provides realistic estimates of event rates in neutrino detectors by folding supernova flux predictions with detector response functions.
- We adopt the Garching supernova model, where the neutrino fluence is characterized by the luminosity, average energy, and a pinching parameter that accounts for deviations from a pure thermal spectrum.
- A 40 kton liquid argon far detector corresponding to the DUNE configuration.
- The energy window is chosen as 1–100 MeV, which covers the relevant supernova neutrino spectrum.
- Three dominant detection channels are included:
  - ν_e Charged-Current (CC) on Argon: $\nu_e + {}^{40}\mathrm{Ar} \to e^- + {}^{40}\mathrm{K}^*$, providing direct sensitivity to electron neutrinos.
  - $\bar{\nu}_e$ Charged-Current (CC) on Argon: $\bar{\nu}_e + {}^{40}\mathrm{Ar} \to e^+ + {}^{40}\mathrm{Cl}^*$, allowing the detection of electron antineutrinos.
  - Elastic Scattering on Electrons: $\nu_\alpha + e^- \to \nu_\alpha + e^-$, which is sensitive to all flavors, though with a strong directional dependence favoring electron neutrinos.

[3] http://www.phy.duke.edu/ schol/snowglobes, "Github," (2021).

Crosssection

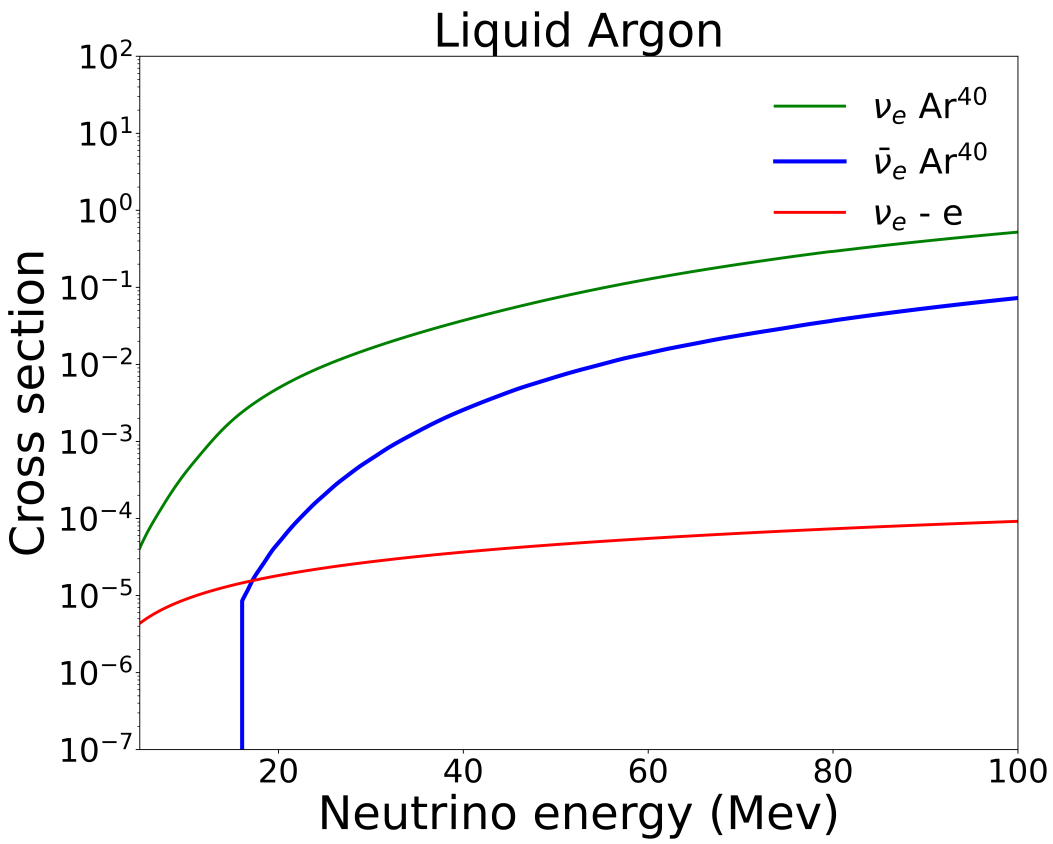


Figure: Cross section for three channels in liquid Argon.

Fluence

- The primary neutrino spectra at the supernova source are modeled as

$$\Phi_\nu(E_\nu) = N\left(\frac{E_\nu}{\langle E_\nu\rangle}\right)^\alpha e^{-(\alpha+1)\frac{E_\nu}{\langle E_\nu\rangle}}, \quad N = \frac{(\alpha+1)^{\alpha+1}}{\langle E_\nu\rangle\Gamma(\alpha+1)}, \tag{14}$$

with total flux

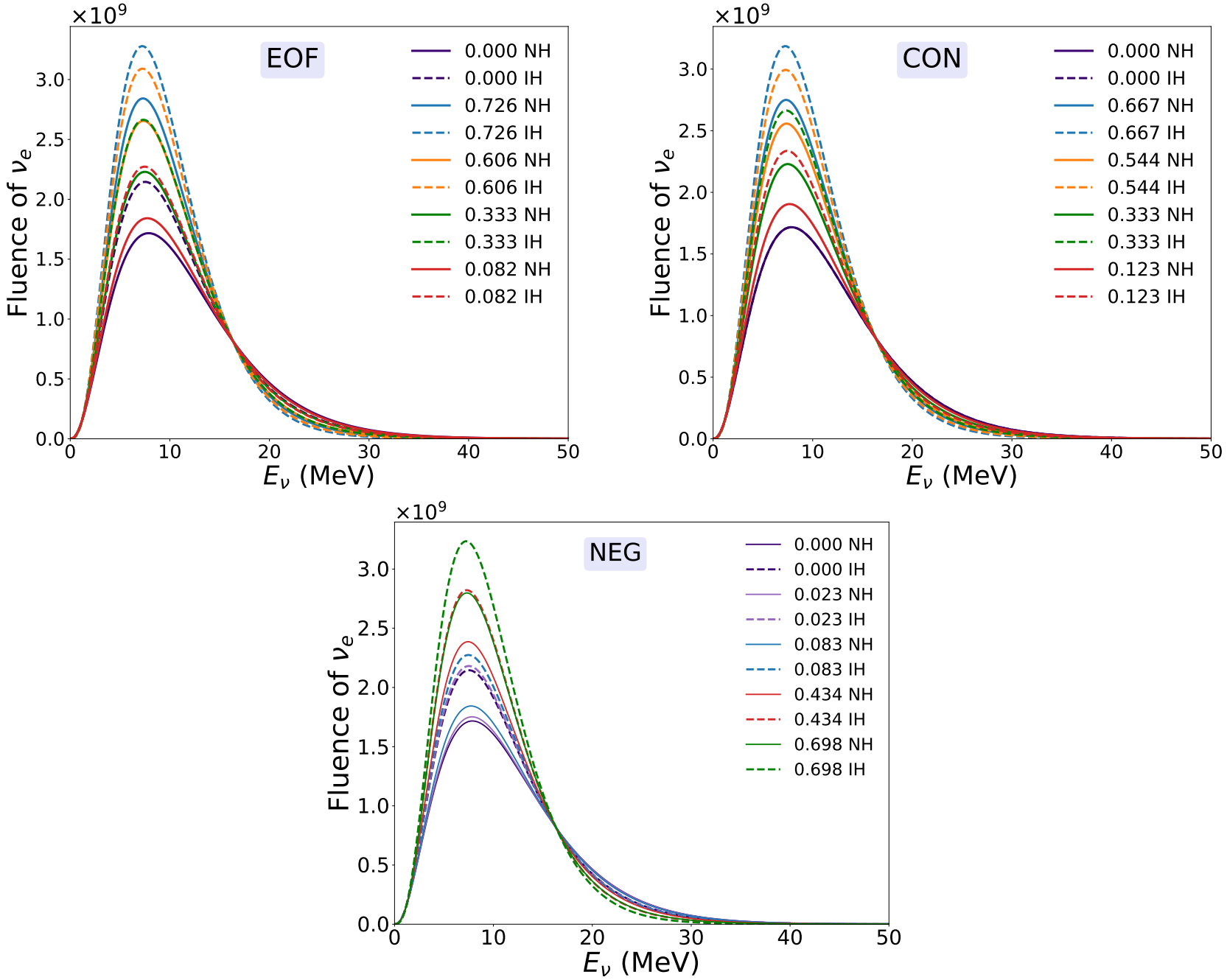
$$F_\nu = \frac{L_\nu}{\langle E_\nu\rangle}\Phi_\nu(E_\nu)\,, \tag{15}$$

where L_ν is the luminosity of the ν type neutrinos.

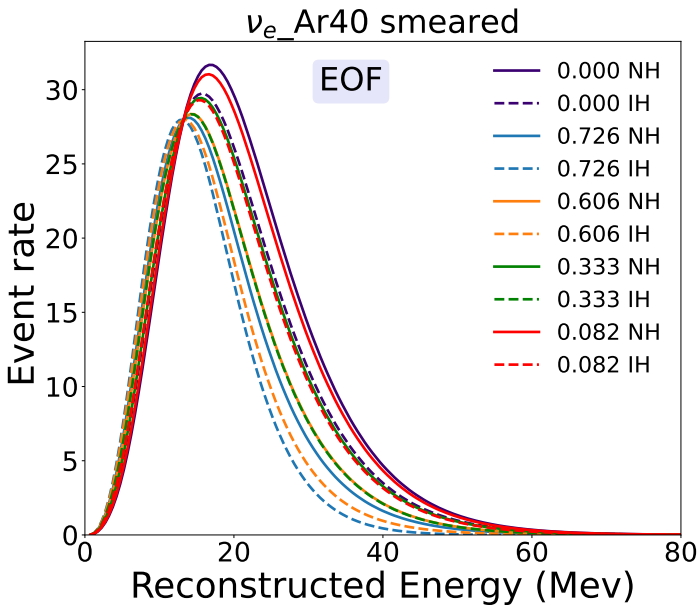
- After applying the effect of neutrino oscillation inside the star's core and surface, the modified fluxes can be defined as

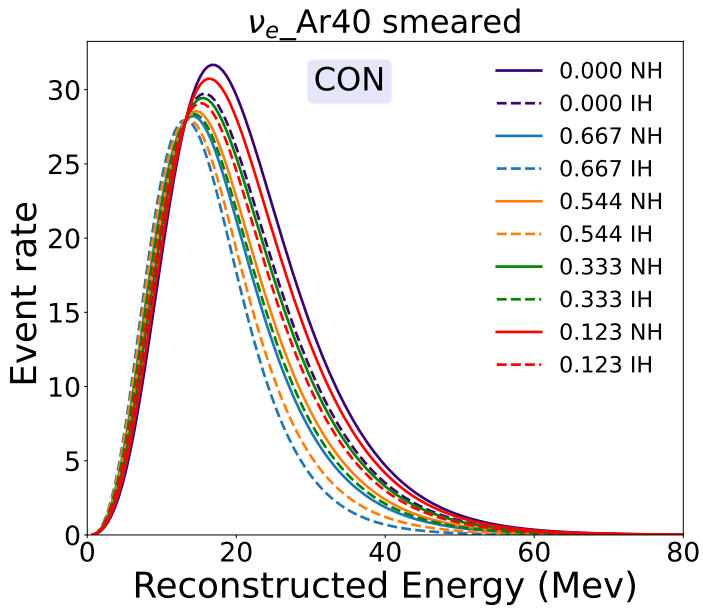
$$\begin{aligned}
F_{\nu_e} &= pF^0_{\nu_e} + (1-p)F^0_{\nu_x}\ ,\\
F_{\bar{\nu}_e} &= \bar{p}F^0_{\bar{\nu}_e} + (1-\bar{p})F^0_{\nu_x}\ ,\\
2F_{\nu_x} &= (1-p)F^0_{\nu_e} + (1+p)F^0_{\nu_x}\ ,\\
2F_{\bar{\nu}_x} &= (1-\bar{p})F^0_{\bar{\nu}_e} + (1+\bar{p})F^0_{\bar{\nu}_x}\ ,
\end{aligned}$$

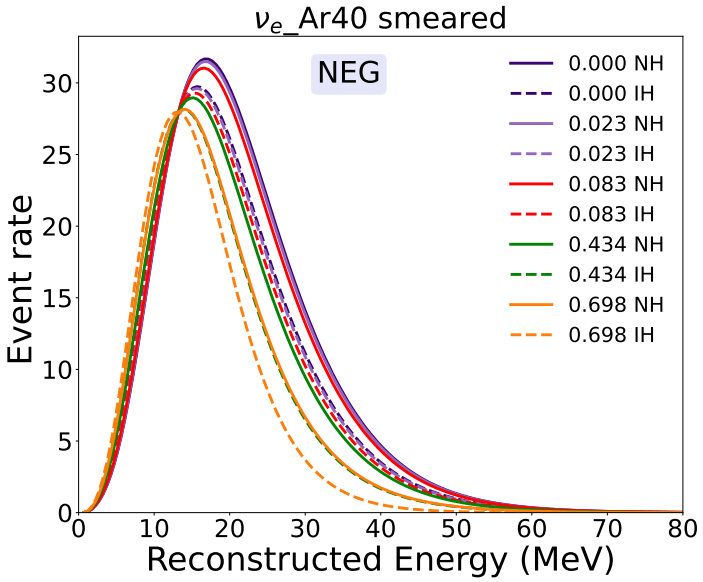
where p and $\bar{p}$ being the survival probabilities for ν_e and $\bar{\nu}_e$ respectively.


EOF
0.000 NH
0.000 IH
0.726 NH
0.726 IH
0.606 NH
0.606 IH
0.333 NH
0.333 IH
0.082 NH
0.082 IH
CON
0.000 NH
0.000 IH
0.667 NH
0.667 IH
0.544 NH
0.544 IH
0.333 NH
0.333 IH
0.123 NH
0.123 IH
NEG
0.000 NH
0.000 IH
0.023 NH
0.023 IH
0.083 NH
0.083 IH
0.434 NH
0.434 IH
0.698 NH
0.698 IH
×10^9
Fluence of ν_e
E_ν (MeV)

Event rate


ν_e_Ar40 smeared
EOF
0.000 NH
0.000 IH
0.726 NH
0.726 IH
0.606 NH
0.606 IH
0.333 NH
0.333 IH
0.082 NH
0.082 IH
Event rate
Reconstructed Energy (Mev)


ν_e_Ar40 smeared
CON
0.000 NH
0.000 IH
0.667 NH
0.667 IH
0.544 NH
0.544 IH
0.333 NH
0.333 IH
0.123 NH
0.123 IH
Event rate
Reconstructed Energy (Mev)

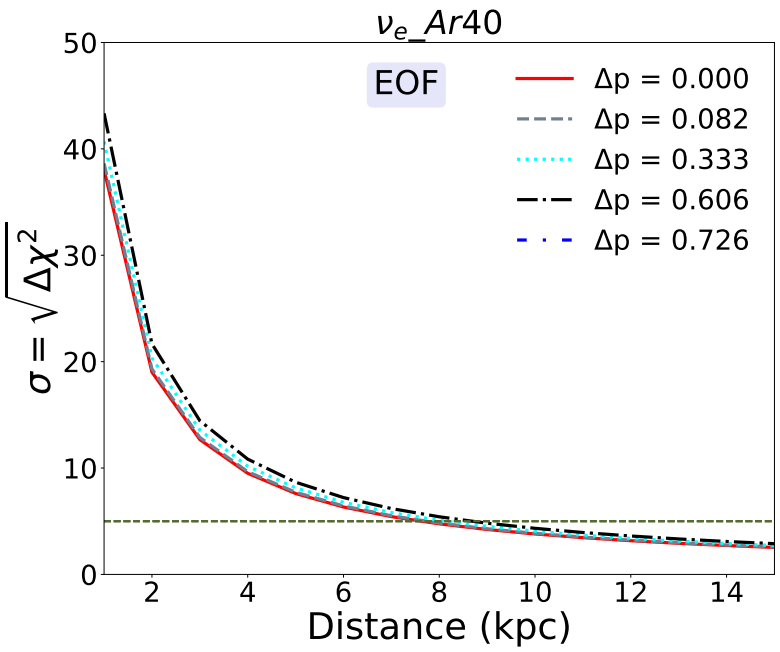

ν_e_Ar40 smeared
NEG
0.000 NH
0.000 IH
0.023 NH
0.023 IH
0.083 NH
0.083 IH
0.434 NH
0.434 IH
0.698 NH
0.698 IH
Event rate
Reconstructed Energy (MeV)

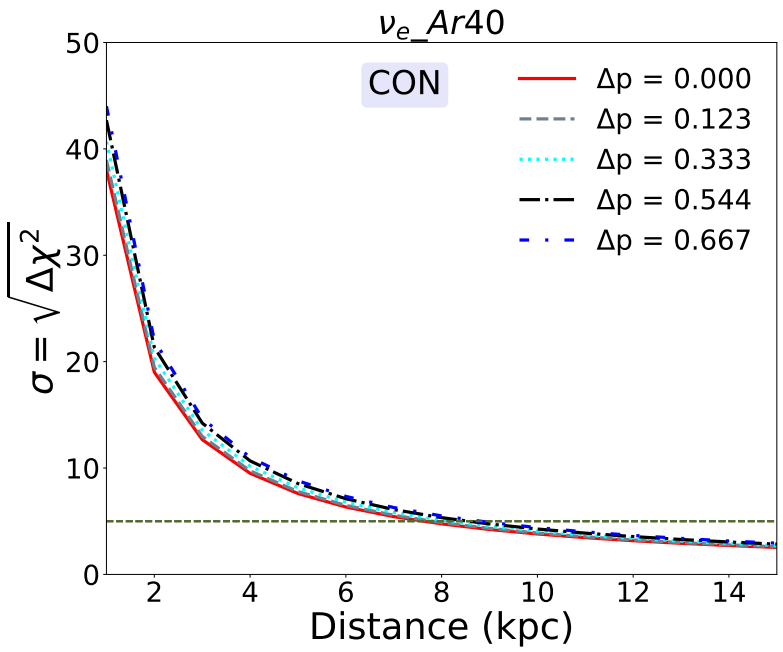
Mass Ordering Sensitivity

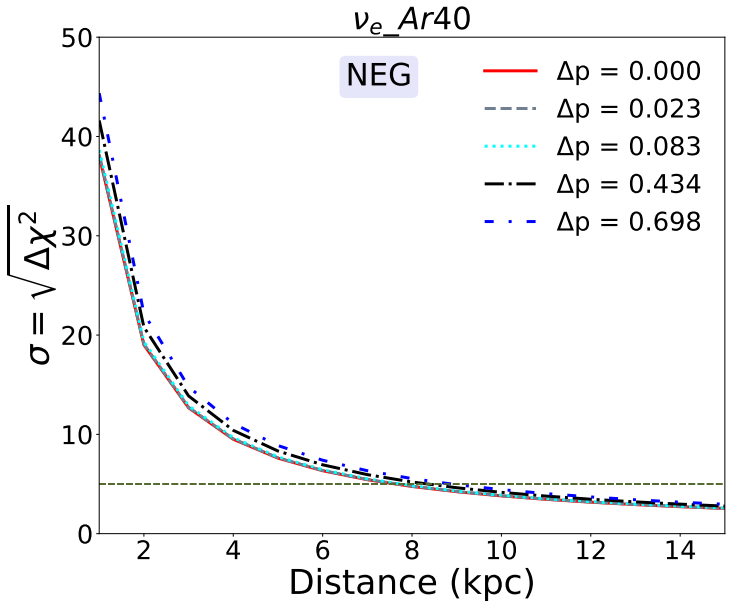
- To quantify the sensitivity to the neutrino mass ordering we employ a Poissonian log-likelihood χ^2 (statistical part), comparing the predicted event spectrum for the test hierarchy with the true-hierarchy spectrum:

$$\chi^2_{\mathrm{stat}} = 2\sum_{i=1}^{n}\left[N_i^{\mathrm{test}} - N_i^{\mathrm{true}} - N_i^{\mathrm{true}}\ln\left(\frac{N_i^{\mathrm{test}}}{N_i^{\mathrm{true}}}\right)\right], \tag{16}$$

where N_i^{true} and N_i^{test} are the predicted event counts in the i-th reconstructed-energy bin for the true and test ordering, respectively, and the sum runs over all bins.


ν_e_Ar40
EOF
Δp = 0.000
Δp = 0.082
Δp = 0.333
Δp = 0.606
Δp = 0.726
$\sigma = \sqrt{\Delta\chi^2}$
Distance (kpc)


ν_e_Ar40
CON
Δp = 0.000
Δp = 0.123
Δp = 0.333
Δp = 0.544
Δp = 0.667
$\sigma = \sqrt{\Delta\chi^2}$
Distance (kpc)


ν_e_Ar40
NEG
Δp = 0.000
Δp = 0.023
Δp = 0.083
Δp = 0.434
Δp = 0.698
$\sigma = \sqrt{\Delta\chi^2}$
Distance (kpc)

Conclusion

- By incorporating neutrino entanglement (EOF, Concurrence, Negativity), we compute updated benchmark predictions for the survival probabilities of supernova neutrinos.
- Entanglement is found to significantly influence the fluence, event rates, and cross-sections across the three dominant detection channels.
- Our mass-ordering sensitivity study shows that a 5σ determination of the neutrino mass ordering is achievable for a core-collapse supernova occurring within a distance of 8 kpc.

Thank You!

Questions?